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Summary

Computer Science and Business student working the full stack of a machine learning product: research and training, the services that run inference, and the front ends on top. Kaggle top 2%; runs a graph neural network on live market data.

Experience

Machine Learning Intern

Jun 15 – Jul 21, 2026

KeyPoint Technologies (Xplore)

- Sole developer of the language model and the full iOS keyboard extension: researched, trained, and evaluated a family of **language models from 4M to 11M parameters** as a drop-in replacement for the product's existing autocorrection and next-word suggestion engine, owning the pipeline end to end from data preparation through **Core ML** conversion and on-device benchmarking.
- Shipped a **4-layer, 256-hidden** model, int8-quantized, that prunes a trie-constrained beam search; improved autocorrection while holding the extension inside the **66MB** limit iOS enforces, cutting the baseline to **15MB** by memory-mapping the 60k-word dictionary trie and n-gram tables into evictable pages.

Software Developer

Dec 2022 – Mar 2023

VHR Infra

- Sole developer; designed, built, and shipped the company website end to end with **Next.js** and **Firebase**.
- Built a logistic regression classifier (scikit-learn) to predict whether prospective clients would accept a proposed price, and prototyped a house-price regression model on client data.

Founding Member and CTO

Dec 2021 – Jun 2022

Y2C (student-founded NGO)

- Sole developer and technical lead; shipped the org's website with **Next.js** SSR on **Vercel**, **Firebase** for data and auth.
- Technical point person from architecture to deployment; represented the org's technology to Y Combinator contacts and NGO backers.

Projects and Competitions

Graph Neural Network Intraday Options Trading — Live Portfolio

Graph neural network that jointly embeds multiple stocks and learns their cross-stock relations, applied in real time to intraday options trading via the **Zerodha Kite API**; live portfolio, peak single-day return **220%** (max drawdown 60%).

Triver — Personal Project

2021 – Present

Preference-first travel recommendations spanning **SVD++**, **Seq2Seq**, and **graph neural networks**; deployed the end-to-end ML pipeline to the web with automated training-to-deployment and production monitoring (MLOps).

Reinforcement Learning — Implementations

Implemented the World Models architecture (VAE, MDN-RNN, evolved controller) on CarRacing; also **PPO**, **SAC**, **DDPG**, and curiosity-driven exploration (ICM) from the original papers, plus GANs for generative modelling.

Stable Diffusion Image-to-Prompts — Kaggle

Finished **Top 2%** of a \$50,000 competition; fine-tuned a CLIP image encoder with an Efficient Channel Attention head to regress Stable Diffusion prompt embeddings on a cosine-similarity objective, and explored Q-Former based approaches.

BirdCLEF 2026 — Kaggle

Finished **Top 10%** of a \$50,000 competition; used an equiangular tight frame (ETF) constrained embedding space to identify few-sample bird species from field audio.

ConserVision Wildlife Image Classification

Finished **15th**; built species classification robust to camera-trap noise: low light, occlusion, and cropped frames.

Skills

Languages: Python, Java, Swift, Kotlin, Go, JavaScript / TypeScript, PHP, MATLAB, CUDA

Machine Learning: PyTorch, TensorFlow, JAX, Flax, scikit-learn, XGBoost, Hugging Face Transformers

Training and Inference: LoRA / PEFT, SFT and DPO, DeepSpeed, FSDP, Ray, vLLM, ONNX Runtime, TensorRT, Core ML, quantization, distillation

MLOps: Weights & Biases, MLflow, Airflow, Docker, Kubernetes

Web and Mobile: React, React Native, Next.js, SwiftUI, micro frontends, Firebase, Vercel

Systems and Data: microservices, Kafka, NATS, Ejabberd, PostgreSQL, TimescaleDB, Elasticsearch, MongoDB, Redis

Education

Two concurrent undergraduate degrees, permitted under UGC's 2022 dual-degree policy (one online, one on campus).

B.Sc. Computer Science — Birla Institute of Technology and Science (BITS), Pilani, Rajasthan 2025 – 2028
Online Degree Programme.

BBA, Artificial Intelligence and Data Science — ICFAI, Hyderabad; full-time, on campus 2024 – 2027

High School, Indus International School, Hyderabad

Grade 10, Delhi Public School, Hyderabad